

Technical specifications

- **PH sensor (4 in 1 pH Sensor Module)**
 - Operating Voltage: 3.3V – 5V DC
 - Output Type: TTL Digital Signals
 - Interface: TTL UART
 - Baud Rate: 9600
 - Length: ~ 40 – 60 mm
 - Width: ~ 30 – 45 mm
 - Weight: 30 grams
 - pH Sensor, LDR Light Sensor, Temperature Sensor (NTC), Water Level Sensor
- **TDS sensor**
 - Operating Voltage: DC 3.3 V – 5.5 V
 - Output Type: Analog voltage (0 – 2.3 V) proportional to TDS value
 - Working Current: ~3 – 6 mA
 - Probe Type: Waterproof TDS probe (2-pin electrode)
 - Probe Cable Length: ~83 cm
 - Module Size: ~29.5 × 36 mm
 - Weight: 100 grams
- **Water Temperature sensor**
 - Sensor Type: Digital temperature sensor (DS18B20)
 - Cable Length: ~ 1 meter (approx. 91 cm) pre-wired
 - Probe Diameter: ~ 6 mm stainless steel tube
 - Probe Length: ~ 30 mm stainless steel section (excluding cable)
 - Operating Voltage: 3.0 V – 5.5 V DC (works with 3.3 V or 5 V logic)
 - Current consumption: 1.5mA
 - Interface: 1-Wire digital communication (one data wire + GND + VCC)
 - Number of Wires: 3 (Red → VCC, Black → GND, White → Data)
 - Temperature Measurement Range: –55 °C to +125 °C
 - Typical Accuracy: ~ ±0.5 °C over much of the range
 - Weight: 30 grams
- **Temperature and humidity sensor (DHT22)**
 - Temperature range: –40 °C to +80 °C
 - Humidity range: 0 % to ~99.9 %
 - Operating voltage: 3.3 V – 5.5 V DC
 - Communication: Digital output using a single-wire data protocol
 - current: Low (~2.5 mA during measurement)
 - L*W*H: 40*23*10
 - Weight: 10g

- **Compass**

- Module Name: GY-271
- Axes: 3-Axis (X, Y, Z magnetic field sensing)
- Communication Interface: I²C (serial communication)
- I²C Address: 0x1E (standard for HMC5883L)
- Resolution: 12-bit ADC (good heading resolution)
- Measurement Range: ±1.3 to ±8.1 Gauss (configurable)
- Operating Voltage: 3 V – 5 V DC (compatible with 3.3 V and 5 V logic)
- Power Consumption: Low power (depends on I²C activity)
- Sampling Rate: Up to ~160 Hz (maximum sensor sample output)
- L*W*H: 15*14*4
- Weight: 3 grams
- Current: 100 micro amp

Kilogram force (Kgf) – pushing force

$$1 \text{ kgf} = 9.81 \text{ Newtons (N)}$$

To pull 4 kg of weight required thrust is 25N

$$\frac{25}{9.81} = 2.55 \text{ kgf}$$

Ideal momentum theory formula:

$$\text{Power} = \frac{T^{3/2}}{\sqrt{2\rho A}}$$

T = Thrust

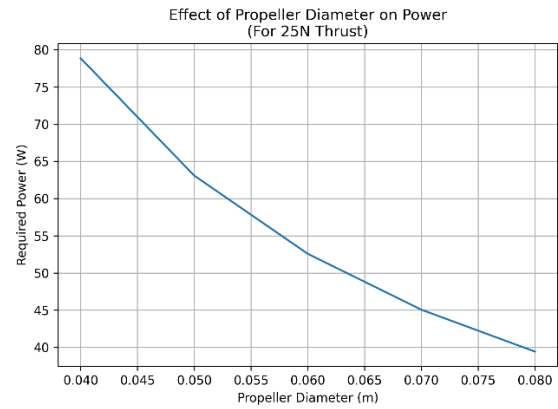
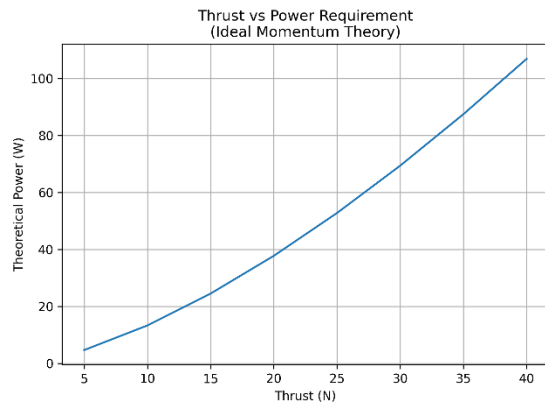
ρ = Water density - 1000 kg/m^3

A = Propeller disk area (m^2) = 60 mm = 0.06 m

$$\text{Area} = \pi r^2 = 3.1416 * (0.03)^2 = 0.0028 \text{ m}^2$$

$$\text{Power} = \frac{25^{\frac{3}{2}}}{\sqrt{2 * 1000 * 0.0028}}$$

$$P = \frac{125}{2.366} = 53 \text{ watt}$$



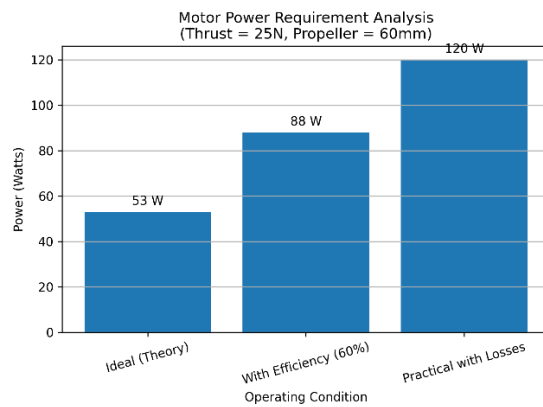
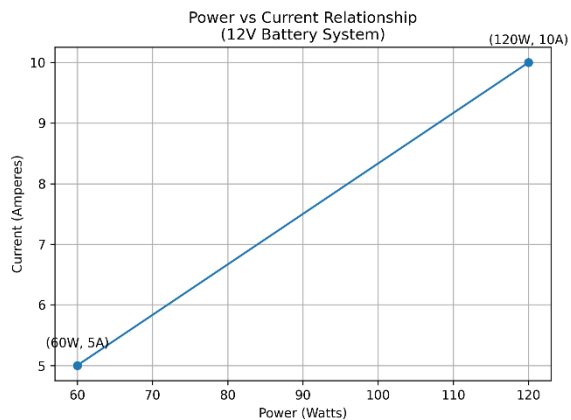
Real world efficiency = $\frac{53}{0.6} = 88 \text{ watt}$

Adding the losses = **100 – 120 watts (60 W *2)**

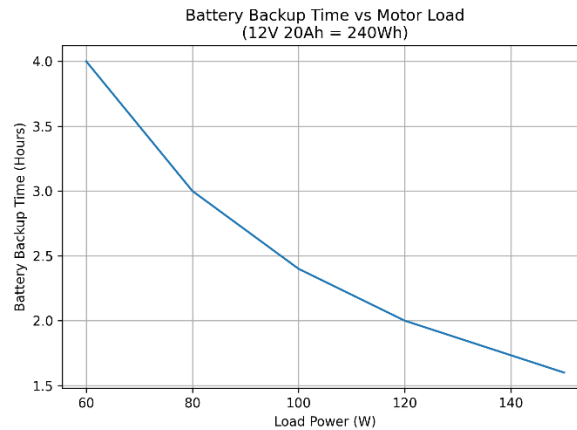
Using two motors for directional control, power per motor drops to 60 watts

Total current requirement for motor:

$$I = \frac{P}{V} = \frac{120}{12} = \mathbf{10 \text{ A}}, \text{ Current per motor} = 5\text{A}$$



Battery and the current requirement = **12V 20Ah battery**



Required motor RPM

Using Actuator Disk Theory (Induced Velocity):

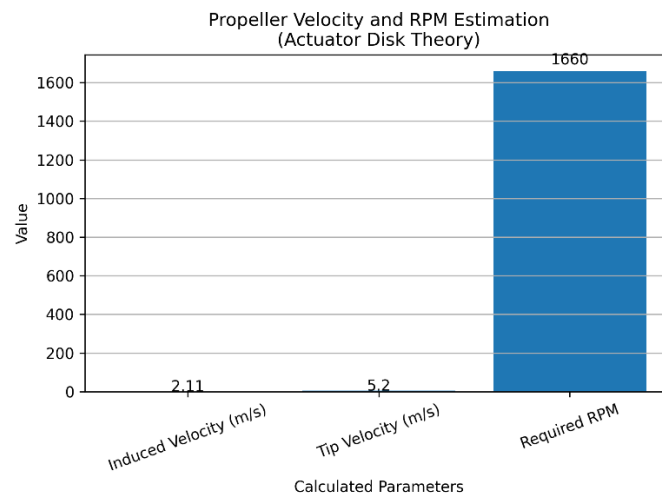
$$V_i = \sqrt{\frac{T}{2\rho A}} = \sqrt{\frac{25}{2*1000*0.0028}} = 2.11 \frac{m}{sec}$$

Water velocity into Propeller RPM:

Tip speed should be 2-3x times of induced velocity: $2.11*2.5 = 5.2 \frac{m}{sec}$

$$V_{tip} = 2\pi r \frac{RPM}{60}$$

$$RPM = \frac{V_{tip} * 60}{2\pi r} = \frac{5.2 * 60}{2*3.1416*0.03} = 1660 \text{ RPM}$$



$$\text{Motor KV Specification} = \frac{2000}{12} = 170 \text{ KV}$$

Electronic speed controller (ESC): 30A ESC for each motor

12v 20Ah Li – Po battery

Solar panel

Solar panel output: $P = V * I$

$$I = 30 / 18 \text{ (Vamp} = 17 - 18\text{V)}$$

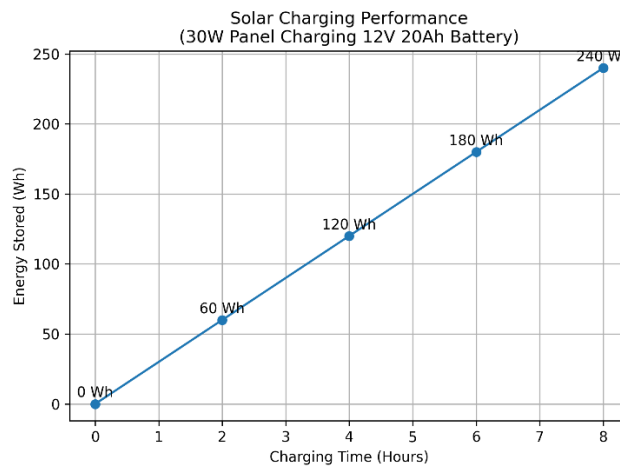
$$I = 1.66 \text{ amp}$$

Battery Watt – hours:

$$12 * 20\text{Ah} = 240\text{Wh}$$

Charging time:

$$\text{charging time: } \frac{\text{Battery Wh}}{\text{Panel W}} = \frac{240}{30} = 8 \text{ hours}$$



Solar panel current output: 1.6A

Applying safety margin:

$$\text{Controller} \geq 1.25 \times \text{Panel current}$$

10A PWM controller for solar panels charging.

References

1. Propeller Theories - Momentum Theory, NPTEL Archive.

<https://archive.nptel.ac.in/content/storage2/courses/101101001/downloads/Intro-Propulsion-Lect-29.pdf>

2. The Actuator Disk Model, Penn State University.

<https://www.e-education.psu.edu/aersp583/node/470>

3. Impeller Tip Speed Definition and Formula, DisperseTech.

<https://www.dispersetech.com/blog/post/tip-speed-definition-and-formula.html>

4. How to Calculate Solar Panel Battery And Inverter, LuxpowerTek.

<https://luxpowertek.com/blog/how-to-calculate-solar-panel-battery-and-inverter/>

5. Solar Panel Charge Time Calculator, HBOWA New Energy.

<https://www.pretapower.com/solar-panel-charge-time-calculator-accurately-estimate-how-long-to-power-your-battery/>